

RESEARCH ARTICLE

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Effect of trap level density on breakdown strength and space charge distribution of polypropylene/low-density polyethylene composites

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Abstract

Polypropylene (PP) as an environmentally friendly insulation material has been paid more attention. It is considered as a potential insulating material to replace cross-linked polyethylene (XLPE) for high voltage direct current (HVDC) applications. In this work, PP and PP/low-density polyethylene (LDPE) composites with different LDPE loading were prepared by melt blending method and the corresponding electrical properties were investigated. Results showed that the addition of LDPE significantly changed the microstructure of PP and the trap density was increased in the composites compared with pure PP. The conduction current of PP/LDPE composites was greater than that of PP. The electrical breakdown strength decreased at low LDPE contents and increased at high contents. Space charge was remarkably suppressed in the blend with 40 wt% LDPE loading. This work indicates that the improvements in the electrical properties, especially the charge transport and space charge suppression in the composites are closely related to the trap level density.

KEYWORDS

composites, polyethylene, polypropylene, space charge, trap level density

1 | INTRODUCTION

Cross-linked polyethylene (XLPE) has been extensively used as bulk insulation material in high voltage direct current (HVDC) power transmission systems due to its excellent electrical and mechanical properties.^[1–4] However, with the continuous development of HVDC transmission technology, the voltage level is increasing and the conductor produces more heat. The use temperature ($\approx 70^\circ\text{C}$) of XLPE hinders its further application. More importantly, XLPE is a kind of thermosetting material, which cannot be recycled when the cable reaches the end of life.^[5–7] Therefore, it is necessary to develop new recyclable insulation materials for HVDC cable applications. Polypropylene (PP) which has high melting

points and excellent electrical properties is a promising material for HVDC cables.^[8,9] However, its stiffness and brittleness at room temperature is not capable as cable insulation material. Some studies have been done to improve the mechanical properties of PP.^[10–12] It was found that the addition of thermoplastic elastomer can effectively improve the mechanical properties of PP material but decrease electrical breakdown strength.^[5,13,14] On the other hand, it has already been proved that the addition of β nucleating agent can enhance the mechanical properties of PP.^[15–19] However, it still cannot meet the requirements of the cable application. Space charge accumulation is another problem that needs to be paid attention to. Polymer dielectric materials can capture charge carriers to form space charge under

high DC voltages which will greatly distort electric field distribution in materials and directly affect insulation aging and breakdown properties.^[20–23] It has become an important restricted factor for HVDC cable development. A large number of studies indicate that the addition of nanoparticles such as SiO₂, MgO, ZnO, Al₂O₃, et al can significantly suppress space charges accumulation in polymer material.^[24–27] Nevertheless, nanoparticles are easy to agglomerate, which can weaken the electrical properties of polymer insulation material. Moreover, it cannot guarantee continuous and stable production in the process of mass production.

Blending can ensure the homogeneity and stability of material compared with nano-doping. Low-density polyethylene (LDPE) is a kind of polyolefin material with a similar structure to PP and the electrical properties of PP/LDPE composites are rarely investigated. In addition, little attention has paid on space charge characteristics of PP/LDPE composites.

In this paper, PP and PP/LDPE composites with different contents of LDPE were prepared. The microstructure, trap level density, space charge, breakdown strength and conduction current were investigated in detail. It is observed that the microstructure of PP has changed after adding LDPE. Meanwhile, the trap level distribution changes with the introduction of LDPE in PP. The space charge distribution is affected by trap level density. When the LDPE content is 40 wt%, the space charge suppression is effectively improved and the breakdown strength increases.

2 | EXPERIMENTAL

2.1 | Materials

PP with a melt flow rate of 3 ± 0.3 g/10 minutes (230°C/2.16 kg) was purchased from Zhongyuan Petrochemical Co., Ltd. The isotacticity of PP is about 98%. LDPE with a melt flow rate of 1.8–2.2 g/10 minutes (190°C/2.16 kg) was provided by BASF-YPC Co., Ltd. The antioxidant 300 was purchased from Yinjing Refinement Technology Co., Ltd., Guang Zhou.

2.2 | Samples preparation

PP and LDPE were dried at 70°C for 24 hours in a vacuum oven before blending. Then, PP and LDPE were blended in high-speed mixer with a rotor speed of 200 rpm for 10 minutes to achieve good dispersion. Finally, PP and LDPE were melt-blended in a Hakke batch mixer at 190°C and the composites with LDPE concentrations of 0, 20, and 40 wt% were prepared. The obtained composites were pressing molded at 200°C under 20 MPa for 10 minutes, and then cooled to room temperature at the same pressure. The obtained films with a thickness of about 200 µm. All the samples were annealed in the vacuum oven at 70°C for 24 hours.

2.3 | Characterization

To observe the microstructure of PP and PP/LDPE composites. The samples were immersed in liquid nitrogen for 2 minutes, then fractured by tweezers to obtain the section surface. After that, the samples were sputter-coated with gold and then imaged by a scanning electron microscope (SEM, JEOL-7001 L).

The polarized optical microscopy (POM) was used to observe the crystal morphology of PP and PP/LDPE composites and the spherulitic size of PP and its composites can be observed by POM on an Axioskop 40 (ZEISS, Germany).

X-ray diffraction (XRD) was carried out on a Panalytical X'Pert PRO XRD (Holland) to characterize the crystal structure of the materials.

Differential scanning calorimetry (DSC, Q2000, America) was performed at temperatures from 30 to 200°C at a heating rate of 10°C/min under a nitrogen atmosphere to study the thermal properties of PP and its composites.

The tensile test was performed according to ASTM D638-2003 using a universal testing machine with a tensile speed 50 mm/min at room temperature. Five specimens for each sample were used for repeat measurements. Dumbbell-shaped specimens with a cross-section of 4 mm × 1 mm and a gauge length of 20 mm were prepared for the measurements.

The trap level pattern of PP and its composites were obtained by thermally stimulated depolarization current (TSDC) measurements and the trap level can be calculated by modified TSDC theory.^[28,29] The experiment procedures are as follows. Before the measurement, the surface of the samples should be clean and then the samples were firstly polarized under $1.25 \text{ kV} \cdot \text{mm}^{-1}$ DC electric field at 50°C for 30 minutes. Then, the samples were cooled to -100°C with a cooling rate of $10^\circ\text{C min}^{-1}$ and stabilized at -100°C for 3 minutes. Then the poling voltage was removed and the samples were short-circuited for 5 minutes. The TSDC was recorded from -100°C to 100°C with a heating rate of 3°C min^{-1} .

The space charge measurement was performed with a pulsed electro-acoustic (PEA) system. All the samples with thickness about 200 µm were under 40 kV mm^{-1} DC electrical field for 40 minutes at 25°C. An electric pulse with an amplitude of 600 V and duration of 5 ns was applied to the samples and the space charge distributions during the polarization process were recorded. Before measurements, all the samples were heat treated at 70°C in the vacuum oven for 24 hours.

DC volume resistivity testing was performed on a three-electrode system (Keithley 6517A). A constant DC electric field of 40 kV/mm was applied on the samples for 1800 seconds at 25°C and the conduction current was recorded.

3 | RESULTS AND DISCUSSION

3.1 | Morphology of the PP and PP/LDPE composites

Figure 1 shows the microstructure of PP and its composites. It can be observed that the PP/LDPE blend form an obvious sea-island structure, where the sea is the PP phase and the islands are LDPE phase when the LDPE content is 20 wt%. In addition, there are many holes which can be seen in the composites, indicating that more interfaces exist between the PP matrix and LDPE in PP/LDPE composites compare to pure PP. With the increase of LDPE content, the morphology of sea-island structure becomes indistinct. This may be due to a similar ratio between PP and LDPE, which makes it difficult to observe the LDPE phase.

3.2 | Crystalline morphology

In order to observe the crystalline morphology of PP and its composites, POM images are shown in Figure 2. It can be seen that the obvious spherulites exist in PP. After adding LDPE, the spherulites size decrease. With the increase of LDPE, the PP crystal is further refined. It may be attributed

to different crystallization temperature of PP and LDPE. PP and LDPE are both crystalline polymer but no eutecticum is formed between them. In addition, PP and LDPE will restrict each other in the crystallization process, which can undermine the spherulitic structure of PP.

3.3 | Crystal structure

The XRD patterns of PP and its composites are presented in Figure 3. It can be seen that the diffraction patterns of the α -crystal with the peaks at $2\theta = 14.0^\circ$, 16.8° , 18.6° , 21.2° , 21.9° , which corresponding to the (110), (040), (130), (111), (131) lattice plane of α crystalline, respectively [30]. The diffraction peak of β -crystal can be observed at $2\theta = 16^\circ$ and 21.2° , corresponding to the (300) and (301) lattice plane of β crystalline. It can also be found that the α (111) and β (301) lattice plane overlap. In addition, the α -crystal and β -crystal co-exist in PP and its composites. With the increase of LDPE content, the diffraction peak intensity at $2\theta = 14.0^\circ$ become weak and the diffraction peak intensity at $2\theta = 21.2^\circ$ become strong. It may be caused by different crystallization behaviors of PP and LDPE. The introduction of LDPE breaks the big spherulitic structure in PP, which is consistent with the result of Figure 2.

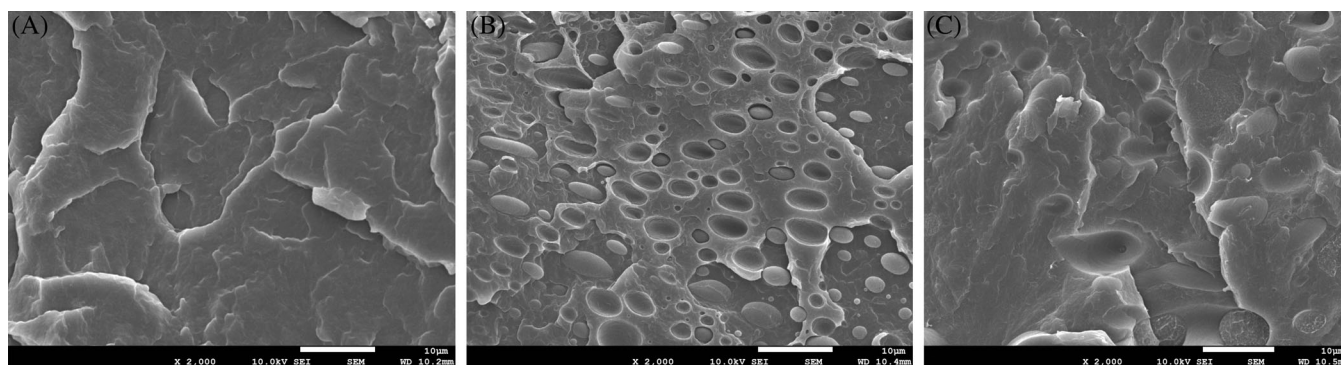


FIGURE 1 SEM images of PP and its composites: A, PP; B, PP/LDPE (80/20); and C, PP/LDPE (60/40). LDPE, low-density polyethylene; PP, polypropylene; SEM, scanning electron microscope

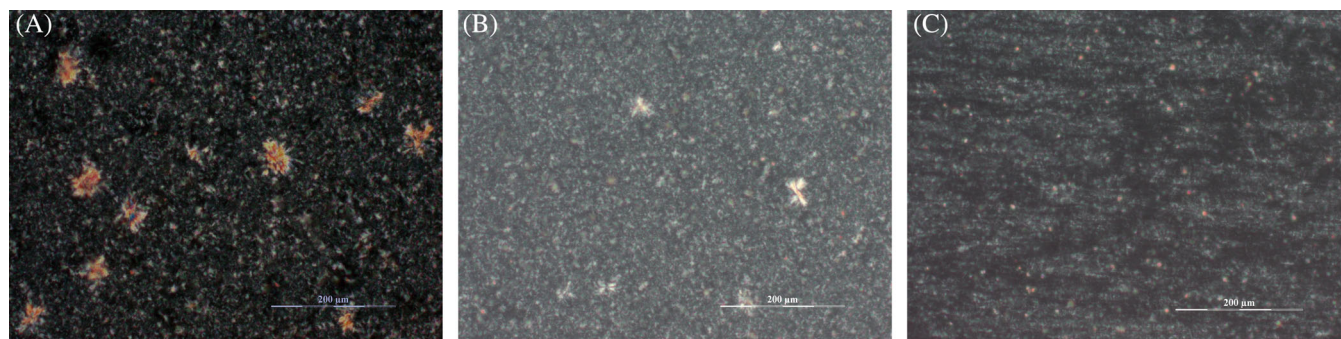


FIGURE 2 POM images of PP and its composites: A, PP; B, PP/LDPE (80/20); and C, PP/LDPE (60/40). LDPE, low-density polyethylene; POM, polarized optical microscopy; PP, polypropylene [Color figure can be viewed at wileyonlinelibrary.com]

3.4 | Thermal properties analysis

Figure 4 shows the melting and crystallization curves of PP and its composites. It can be observed from Figure 4A that the two melting peaks appear in PP due to the existence of α crystalline and β crystalline. After adding LDPE, three melting peaks exist in PP/LDPE composites. It indicates that the PP and LDPE do not have perfect compatible and the PP/LDPE composites have two phases not homogeneous single phase. On the other hand, it suggests the β phase exists in PP/LDPE composites. The melting temperature of PP and LDPE are 161°C and 106°C, respectively. It is found from Figure 4B that the crystallization temperature of PP is higher than that of LDPE and the peak area of PP decrease with the increase of LDPE content. It suggests that the PP and LDPE are crystallized separately, the molecular chain of LDPE does not enter the lattice of PP.

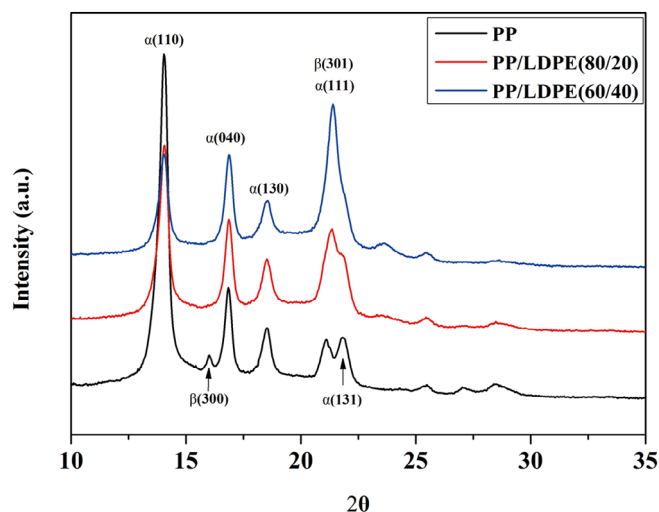


FIGURE 3 XRD patterns of PP and its composites. PP, polypropylene; XRD, X-ray diffraction [Color figure can be viewed at wileyonlinelibrary.com]

3.5 | Mechanical behavior

Mechanical behavior is an important factor for cable insulation application. The typical tensile stress-strain curves of PP and its composites are exhibited in Figure 5. The tensile yield strength and breaking elongation of PP and its composites are listed as an inset in Figure 5. It can be seen that the tensile yield strength decreases and the breaking elongation increases as the LDPE content increases. The decrease of tensile yield strength is mainly attributed to the lower tensile yield strength. LDPE enhance the breaking elongation of PP, which should be attributed to the good mechanical properties of LDPE. On the other hand, the decrease of spherulite size results in an increase of breaking elongation, which is consistent with the results of POM.

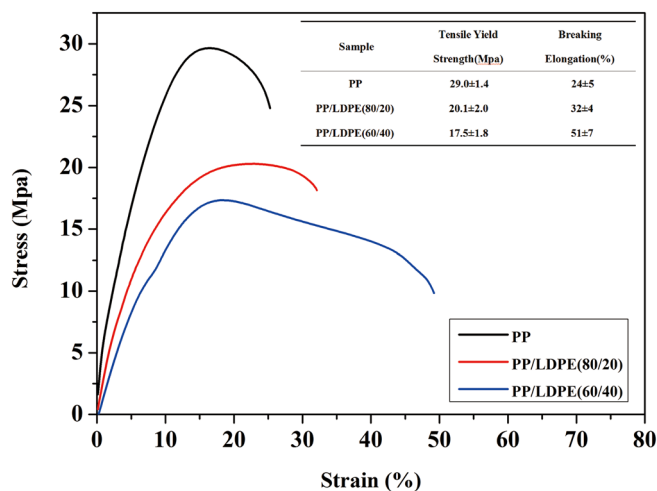


FIGURE 5 Tensile stress-strain curves of PP and its composites at 25°C. PP, polypropylene [Color figure can be viewed at wileyonlinelibrary.com]

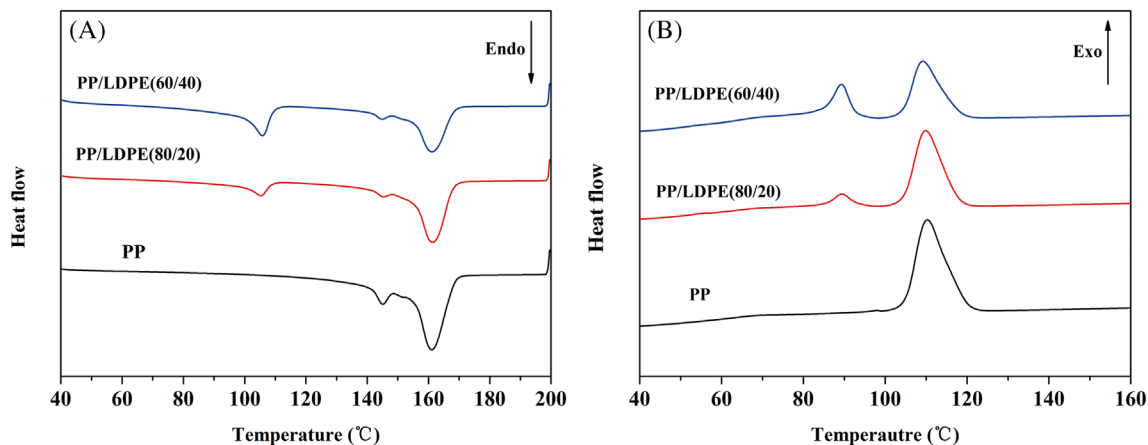


FIGURE 4 A, DSC melting and B, crystallization curves of PP and its composites. DSC, differential scanning calorimetry; PP, polypropylene [Color figure can be viewed at wileyonlinelibrary.com]

3.6 | Trap level distribution

The trap level distributions of PP and its composites are shown in Figure 6. The trap level distribution is calculated from the TSDC spectrum. The results show that the trap level distributions of PP and PP composites are very different. The trap level of PP is in the range of 0.9 to 1.01 eV with a peak density at 0.956 eV, which can be regarded as deep traps in PP compared with the traps in PP/LDPE composites. After adding LDPE, the trap level density significantly increase and the trap level range of PP composites become wider. The peak value of trap density in PP/LDPE (80/20) is about $4.49 \times 10^{19} \text{ m}^{-3} \text{ eV}$ which is 2.6 times the peak value in PP ($1.72 \times 10^{19} \text{ m}^{-3} \text{ eV}$). With the increase of LDPE loading, the trap level density of PP composites decrease. Previous works have proved that the traps were mainly located in the interface.^[31] Therefore, the variation of trap level density is attributed to the generation of interfaces between PP and LDPE.

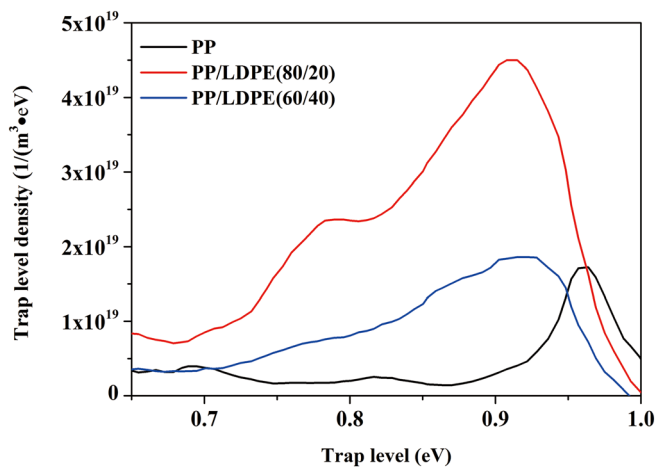


FIGURE 6 Trap level distributions of PP and its composites. PP, polypropylene [Color figure can be viewed at [wileyonlinelibrary.com](#)]

3.7 | Electrical breakdown strength

Figure 7 shows the Weibull distributions of PP and its composites. The AC and DC breakdown strength can be calculated by the following equation^[32]:

$$P = 1 - \exp \left[- \left(\frac{E}{E_0} \right)^\beta \right] \quad (1)$$

where E is the experiment results of breakdown strength, β is the shape parameter and E_0 is the characteristic breakdown strength. E_0 can be obtained when the cumulative breakdown probability is at 63.2%. P is the cumulative probability of breakdown. The probability of breakdown can be written as follows:

$$P_i = \frac{i - 0.44}{n + 0.25} \times 100\% \quad (2)$$

where P_i is the cumulative probability of the breakdown data, i is the result when the breakdown data of E are sorted in ascending order and n is the number of specimens of each sample ($n = 10$ in this work). It can be seen from Figure 7A that the AC breakdown strength of PP/LDPE composites decreases when the LDPE content is 20 wt%, which is consistent with the result of DC breakdown strength from Figure 7B. However, with the increase of LDPE loading, the AC and DC breakdown strength of PP/LDPE composites increase. It is found that the AC breakdown strength of PP/LDPE composites decreases from 107 kV/mm to 90 kV/mm, then increases to 126 kV/mm with the increase of LDPE content. At the same time, the DC breakdown strength of PP/LDPE composites decreases from 182 kV/mm to 173 kV/mm, then increase to 367 kV/mm as the LDPE content increase. The reason for this phenomenon is as follows. For polymer, the free path length of electrons is

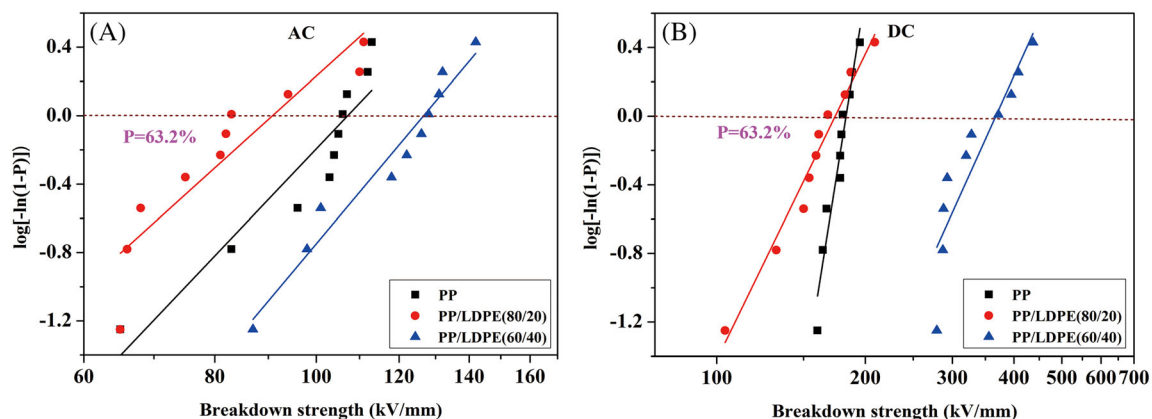


FIGURE 7 Weibull distributions of PP and its composites: A, AC breakdown strength and B, DC breakdown strength. PP, polypropylene [Color figure can be viewed at [wileyonlinelibrary.com](#)]

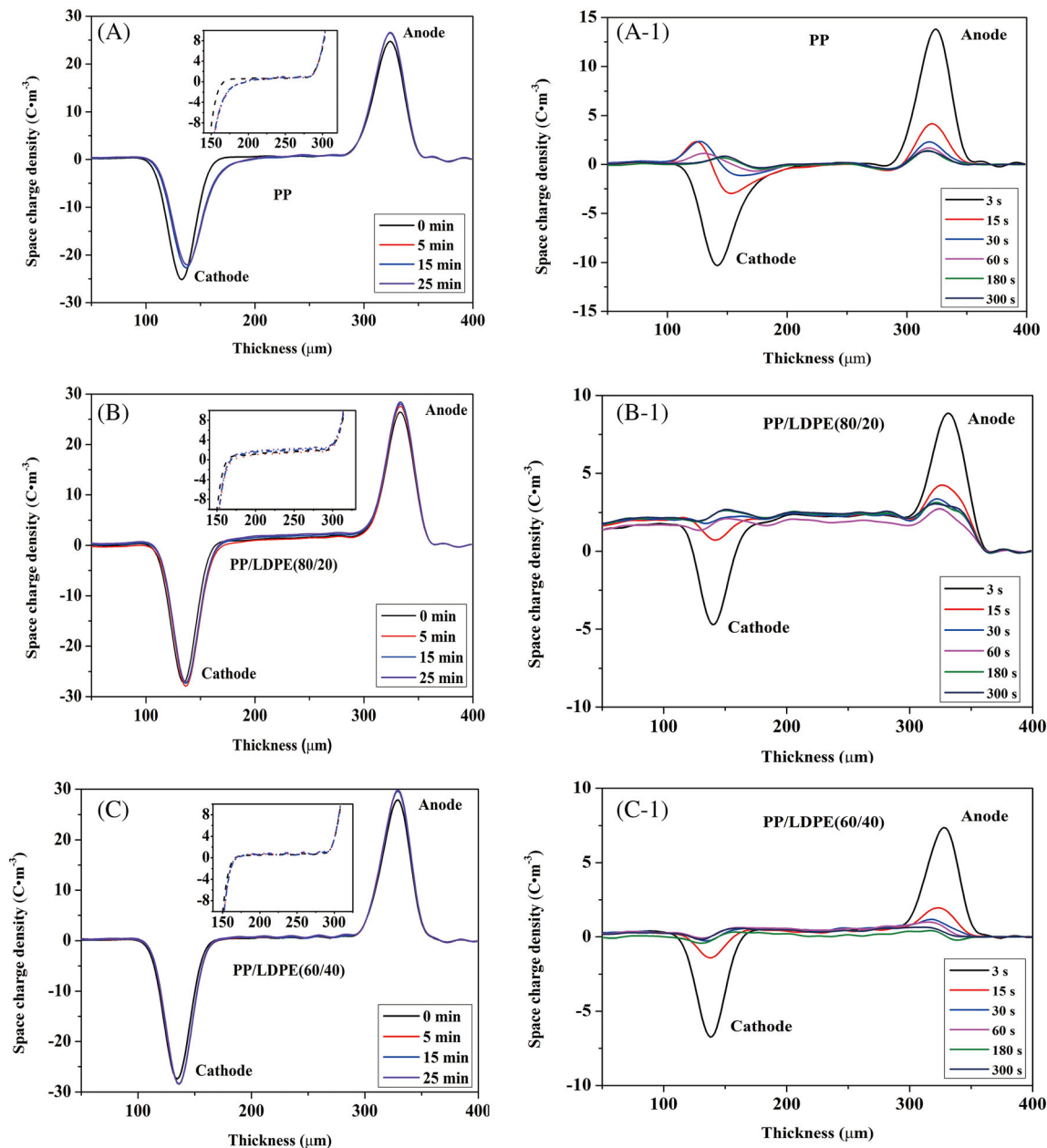


FIGURE 8 Space charge distributions of PP and its composites under 40 kV/mm at room temperature: A, PP; B, PP/LDPE (80/20); and C, PP/LDPE (60/40), corresponding voltage off space charge profiles: A1, PP; B1, PP/LDPE (80/20); and C1, PP/LDPE (60/40). LDPE, low-density polyethylene; PP, polypropylene [Color figure can be viewed at wileyonlinelibrary.com]

short and the injected electrons from the electrode are quickly trapped after a short distance migration. The introduction of LDPE produces a large number of more shallow traps which can contribute to transfer electrons. It increases the electron free path. Thus, the breakdown strength of PP/LDPE composites decreases. It can also be found that the breakdown strength of PP/LDPE composites increases after adding 40 wt% LDPE, which can be attributed to the variation of the trap level density distribution. When the content of LDPE reaches to 40%, the trap level density distribution become narrower and focus on higher trap level. Therefore,

it can effectively prevent carriers migration, which results in an increase in breakdown strength.

3.8 | Space charge distribution

Figure 8 shows the space charge behaviors of PP and its composites under 40 kV/mm for 25 minutes. It can be seen from Figure 8A that a large number of homo-charges are accumulated near the cathode of PP, which is the result of electron injection at the cathode. This phenomenon may be caused by the interfaces between crystalline and amorphous

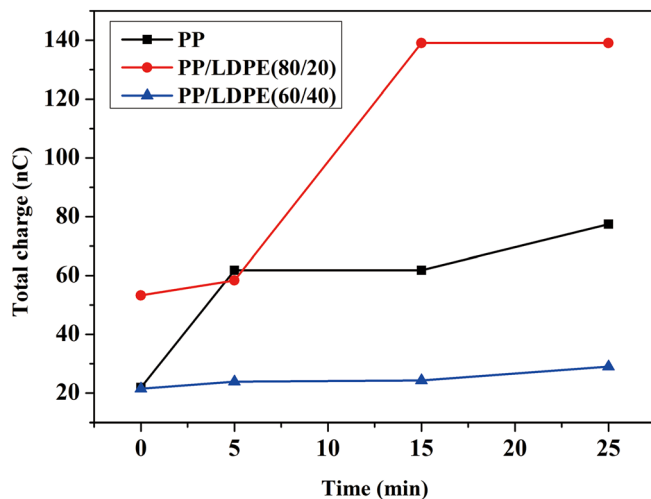


FIGURE 9 The total absolute amount of charge accumulated in PP and its composites. PP, polypropylene [Color figure can be viewed at [wileyonlinelibrary.com](#)]

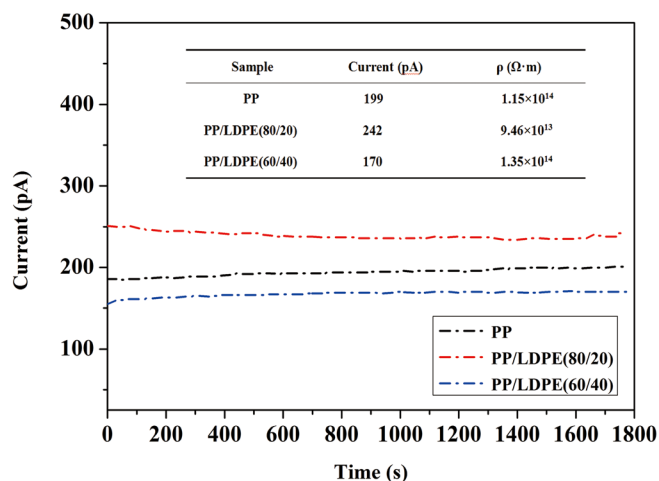


FIGURE 10 Conduction current curves of PP and its composites. PP, polypropylene [Color figure can be viewed at [wileyonlinelibrary.com](#)]

regions in PP. The interfaces can introduce trapping sites which can capture injected charges. When 20 wt% LDPE was added to PP, the space charge distribution in PP changed as can be seen from Figure 8B. A certain amount of charges are uniformly distributed in PP/LDPE, which is related to the interfaces between PP and LDPE. The interface means the addition of LDPE produces a large number of shallow traps in PP, which can be found from the results of trap level distribution. Shallow traps make it easier for charges to migrate in PP. With the increase of LDPE content, there are almost no space charges existing near the cathode and anode. Only a small amount of space charges were observed inside PP. Moreover, the total absolute amount of charge accumulated in the samples can be seen in

Figure 9. The results show that the application time of DC voltage increase, the amount of total charge increase. However, when the content of LDPE is 40 wt%, the variation of charge tends to be stable.

To evaluate the effect of space charge that captured by traps, the space charge profiles of PP and its composites at short circuit condition are shown in Figure 8A1-C1. It can be seen from Figure 8A1 that the space charge dissipation is slow near the cathode and polarity reversal occurs. It indicates that the injected charges are captured by deep traps. The introduction of LDPE changes the dissipation characteristics of charge in PP. It is found from Figure 8B1,C1 that the rate of charge dissipation becomes faster. It may be attributed to the changes in trap level and trap level density.

3.9 | Conduction current

The conduction current was measured to evaluate the insulation properties. The conduction current curves of PP and its composites are shown in Figure 10. The volume resistivity is calculated by current value, which is the average of 1700 seconds to 1800 seconds. The volume resistivity of PP is $1.15 \times 10^{14} \Omega \text{ m}$. When LDPE is added to PP, the volume resistivity of PP/LDPE (80/20) and PP/LDPE (60/40) are $9.46 \times 10^{13} \Omega \text{ m}$ and $1.35 \times 10^{14} \Omega \text{ m}$, respectively. From the results, it can be seen that the volume resistivity of PP decreases after adding LDPE. It is mainly due to the variation of trap level density. A large number of shallow traps make it easier for carriers to migrate in PP composites. The electrons can get out easily from swallow traps. Therefore, the volume resistivity of PP decreases. In addition, the volume resistivity of PP/LDPE (60/40) is higher than that of PP/LDPE (80/20). It is related to the decrease of shallow traps density, which makes it difficult for electrons to migrate.

4 | CONCLUSION

In conclusion, LDPE was dispersed uniformly in the PP matrix and a typical sea-island structure was formed. The sea-island structure became blurred with the increase of LDPE content. The spherulite size of PP decreased gradually with the increase of LDPE content. The space charge distribution was obviously influenced by the trap level density. The interfaces between PP and LDPE can form shallow traps compared with the traps in pure PP, where the shallow traps were mainly located at the interface areas. It made space charge migration become easier through these shallow traps. The addition of LDPE increased the trap level density in PP, which can change the space charge distribution in PP composites. Thus the space charge accumulation has been significantly improved. Meanwhile, the breakdown strength and conduction current were also mainly related to the change of trap level density.

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